QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2026
FIFTH SEMESTER
SEC-MG5SECBIF300 – INTRODUCTION TO R PROGRAMMING
(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks:** 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)
Students should attempt at least one question from each Course Outcome to enhance their overall outcome attainability.

Part A – Short Answer Questions

Answer any 5 questions. Each question carries 2 marks.

1. What is RStudio? Mention one advantage of using RStudio for R programming. [U][1]

2. List any four basic data types in R. [U][2]

3. What are vectors in R? Give one example of creating a vector. [U][3]

4. What is the purpose of install.packages() in R? [U][4]

5. Differentiate between a data frame and a matrix in R. [U][5]

6. What is ggplot? Mention one type of graph that can be created using it. [U][6]

7. Write the R command used to read a CSV file. [A][5]

[5 × 2 = 10]

Part B – Short Essay Questions

Answer any 3 questions. Each question carries 5 marks.

8. Explain R operators and operator precedence with suitable examples. [U][2]

9. Explain decision-making statements in R with examples. [A][1]

10. Describe lists, data frames and matrices in R, highlighting their major characteristics. [U][3]

11. Explain dot charts, bar plots and box plots in R. Mention one use of each. [A][6]

12. Explain the procedure for reading and writing CSV/Excel and other files using R. [A][5]

[3 × 5 = 15]

Part C – Long Essay Questions

Answer any 1 question. Each question carries 10 marks.

13. Explain R data structures in detail. Discuss vectors, lists, data frames, matrices, arrays and factors with suitable examples. [A][3]

14. Explain data visualization in R. Discuss the R plot function and ggplot, and describe bar plots, box plots, pie charts, scatter plots, line charts and histograms with suitable applications. [S][6]

15. Explain loops and functions in R in detail. Illustrate their use with suitable R programming examples and discuss how they support programming logic. [E][4]

[1 × 10 = 10]